Phishing Email Analysis Report

Task: Cyber Security Internship, Task 2 – Analyze a Phishing Email Sample **Analyst:** Shuvo Paul **Date of analysis:** 8 November 2026 **Mailbox:** Gmail (shuvopaul601@gmail.com) **Tools used:** WhatIsMyIP.com Email Header Analyzer, Sendmarc Email Header Analyzer, manual review of raw headers and HTML source

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1. Objective

Examine two emails received in Gmail, analyse their headers, sender, links and content, and identify phishing indicators. One email is suspicious, the other is used as a legitimate comparison.

#	Sender	Subject	Verdict
1	Elisabeth `<elisabeth@flexeni.co>`	Come steady the thought she keeps having	**Malicious / phishing-style lure (high risk)**
2	`no-reply@sampark.gov.in`	Happy Diwali to You!	**Likely legitimate (low risk)**

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2. Email 1 – `elisabeth@flexeni.co` (Suspicious)

2.1 Header summary

Field	Value
From	Elisabeth elisabeth@flexeni.co
To	shuvopaul601@gmail.com
Reply-To	elisabeth@flexeni.co
Return-Path	bounce@flexeni.co
Subject	Come steady the thought she keeps having
Date	Sun, 8 Nov 2026 07:35:28 +0100 (received 7 Nov 2026 22:35 PST)
Message-ID	f2eb7ee8e968730fa1300d8016b45c45@iju.flexeni.co
Sending server	course-prime177.flexeni.co
Origin IP	195.178.129.187
Location / ISP	Kyiv, Ukraine – URAN Network (ASN 3319)
Hops	2 (sender server → Google MX, TLS 1.3)
SPF / DKIM / DMARC	pass / pass / pass (DMARC policy: quarantine)

2.2 Phishing indicators found

#	Indicator	Evidence	Risk
1	**Unsolicited, emotionally manipulative lure**	Body is written as if about a "she" who is "bored, restless" and wants "company"; headline says she needs "a man who can keep up". This is a classic romance/adult-bait social engineering hook.	High
2	**Vague, odd subject line**	"Come steady the thought she keeps having" is not a normal business or personal subject; it is designed to provoke curiosity.	Medium
3	**Fake persona / generic sender**	A first-name-only sender ("Elisabeth") with no surname, no relationship to the recipient, and a signature using emojis (candle, wine, fire) to build intimacy.	High
4	**Single call-to-action button**	The whole email pushes one button: **"Keep My Mood Company"**. Nothing else in the mail has any real information.	High
5	**Obfuscated tracking links**	The button points to `https://iju.flexeni.co/c/AWMAAAAA...` – a long encoded redirect URL that hides the final destination. The real landing page cannot be seen from the email.	High
6	**Tracking pixel**	A 1x1 hidden image (`iju.flexeni.co/o/...` , `display:none`) confirms when the mail is opened, so the sender can validate that the address is live.	Medium
7	**Hidden preheader text**	A `display:none; font-size:1px; opacity:0` div contains the text "She is bored, restless, and ready for company…". This is hidden text used to control the inbox preview.	Medium
8	**Empty plain-text part**	The `text/plain` part of the multipart message is empty; only HTML is delivered. This is common in bulk spam and helps evade text-based filters.	Medium
9	**Suspicious server hostname**	The sending host is `course-prime177.flexeni.co` – a numbered "course" host that has nothing to do with the message content. This points to rented or bulk-mail infrastructure.	Medium
10	**Sender infrastructure in Ukraine**	Origin IP 195.178.129.187 geolocates to Kyiv, Ukraine, yet the footer claims a US address. The location mismatch suggests a spoofed or fabricated identity.	High
11	**Fake / misleading postal address**	Footer lists "4510 Booth by Hill Avenue, Aberdeen Proving Ground, MD 21005" – a US Army installation, which is not a plausible commercial mail address. It looks inserted only to appear legal (CAN-SPAM style).	High

#	Indicator	Evidence	Risk
12	**Strange footer wording**	"dim the late note with Unsubscribe" – unnatural phrasing instead of a clear opt-out, typical of templated scam mail.	Low
13	**Weak DKIM algorithm**	DKIM signature uses `rsa-sha1`, which is deprecated and considered weak. The selector is a long random number (`10032623245085196 1225`) rather than a normal named selector.	Medium
14	**Newly-registered-looking domain**	`flexeni.co` (.co TLD) is unknown and not tied to any recognisable brand. Domain age should be checked using WHOIS.	Medium
15	**Odd writing style**	Sentences are intentionally cryptic ("make the quiet behave", "dim the late note") – unnatural grammar and wording, not normal human correspondence.	Medium

2.3 Important note: authentication PASS does not mean safe

SPF, DKIM and DMARC all passed. This only proves the email was really sent by someone who **controls the domain** `flexeni.co`. It does **not** prove the sender is trustworthy. Attackers routinely register their own domains and configure authentication correctly, so the mail passes technical checks while the content is still malicious. Here, the content, infrastructure, address mismatch and link behaviour are the red flags.

2.4 Conclusion for Email 1

Verdict: Phishing / scam solicitation (high risk). It is a mass-mailed social-engineering lure whose goal is to get the recipient to click the tracked button. The probable outcomes are an adult/dating scam site, credential or payment-detail harvesting, or confirming a live address for further spam.

Recommended actions

1. Do not click the button, the unsubscribe link or any other link.
2. Do not reply (confirms the mailbox is active).
3. In Gmail, use **Report phishing**, then delete the message.
4. Block the sender / domain.
5. (Optional) check `flexeni.co` in WHOIS and VirusTotal, and scan the redirect URL in a safe sandbox such as urlscan.io.

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3. Email 2 – `no-reply@sampark.gov.in` (Comparison / likely legitimate)

3.1 Header summary

Field	Value
From	no-reply@sampark.gov.in
To	shuvopaul601@gmail.com
Subject	Happy Diwali to You!
Date	Sat, 7 Nov 2026 14:44:51 +0530 (IST)
Route	localhost → rabbit-mq8.nic.in (10.247.254.118) → vastu156134.nic.in (164.100.156.202, New Delhi) → Google
Delivery time	1 min 58 sec
SPF / DKIM / DMARC	pass / pass / pass (d=sampark.gov.in, s=sampark1)
Sendmarc grade	B (84/100)

3.2 Observations

#	Observation	Assessment
1	Domain is a `.gov.in` government domain, and the mail was routed through `nic.in` (National Informatics Centre) servers in New Delhi.	Consistent with a real Government of India messaging system.
2	SPF, DKIM, DMARC all pass for `sampark.gov.in`, with a properly named selector (`sampark1`).	Good authentication.
3	Origin IP shows `127.0.0.1` and a private `10.x` hop.	Normal for internal mail systems (the application hands the message to a local relay); not a spoofing sign.
4	Sendmarc flagged **"Unencrypted Transmission" (medium)**.	Only the internal hops lacked TLS; the external hop to Google was encrypted. Minor issue, not malicious.
5	Timestamps are consistent (14:44 IST = 09:14 UTC) and the 1m 58s delay is reasonable.	No tampering signs.
6	Content is a seasonal greeting with no request for credentials, money, or urgent action. Diwali 2026 falls on 8 Nov, so the timing fits.	No social-engineering pressure.

3.3 Conclusion for Email 2

Verdict: Likely legitimate (low risk). No spoofing, no mismatched domains and no urgent or threatening language were seen in the header data. Caveat: only header analysis was available for this mail. Links and attachments in the body, if any, should still be hovered over and checked before clicking.

4. Side-by-side comparison

Check	Email 1 (flexeni.co)	Email 2 (sampark.gov.in)
Sender domain	Unknown .co domain, no brand	Official .gov.in domain
SPF / DKIM / DMARC	Pass (attacker-owned domain)	Pass (government-owned domain)
Sending infrastructure	Numbered "course" host, Kyiv, Ukraine	NIC mail servers, New Delhi, India
Location vs claimed identity	Mismatch (Ukraine vs fake US address)	Consistent
Links	Obfuscated tracking redirects, single CTA button	None flagged
Tracking pixel / hidden text	Yes / Yes	Not observed
Language	Cryptic, manipulative, emotional bait	Plain seasonal greeting
Urgency / pressure	Emotional pressure to click	None
DKIM algorithm	rsa-sha1 (weak)	Not flagged
Verdict	**Phishing / scam – High**	**Legitimate – Low**

Key lesson: passing SPF/DKIM/DMARC shows *who* sent a message, not *whether it is safe*. Always combine header results with sender reputation, link inspection and content analysis.

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5. Method (steps followed)

1. Opened each email in Gmail and used **Show original** to copy the full raw headers.
2. Ran the headers through two free analyzers (WhatIsMyIP and Sendmarc).
3. Checked the From / Reply-To / Return-Path for mismatches and spoofing.
4. Reviewed SPF, DKIM and DMARC results and the delivery path (hops, IPs, geolocation).
5. Read the HTML source to find links, hidden elements, tracking pixels and footer details.
6. Assessed language, tone and urgency.
7. Summarised the phishing traits and compared the two emails.

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6. Interview questions – short answers

1. **What is phishing?** A cyber attack where a fake message tricks a victim into revealing data, clicking a malicious link, or installing malware.

2. **How to identify a phishing email?** Check the sender address/domain, unexpected requests, urgent or emotional language, mismatched or hidden links, attachments, and spelling or grammar errors.
3. **What is email spoofing?** Forging the sender details (From address or display name) so the email appears to come from a trusted source.
4. **Why are phishing emails dangerous?** They can lead to credential theft, financial loss, malware or ransomware infection, and data breaches.
5. **How can you verify the sender's authenticity?** Inspect the full headers, check SPF/DKIM/DMARC, verify the domain (WHOIS), and contact the organisation through an official channel.
6. **What tools can analyze email headers?** Google Admin Toolbox Messageheader, MXToolbox Header Analyzer, WhatIsMyIP, Sendmarc, and similar.
7. **What actions should be taken on suspected phishing emails?** Do not click or reply, report it (Gmail: Report phishing), delete it, inform the IT/security team, and change passwords if something was clicked.
8. **How do attackers use social engineering in phishing?** They exploit emotions such as curiosity, fear, urgency, greed or attraction, and pose as trusted people or brands to push the victim into acting without thinking.

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7. Summary

- **Email 1 (flexeni.co):** phishing-style lure with 15 indicators, including an obfuscated tracking link, a fake US address, Ukrainian server infrastructure, hidden text and a tracking pixel. Its passing SPF/DKIM/DMARC results do not make it safe.
- **Email 2 (sampark.gov.in):** consistent with a genuine NIC / Government of India mail with only a minor internal-hop encryption finding.